

Course kick off

Lecture 1

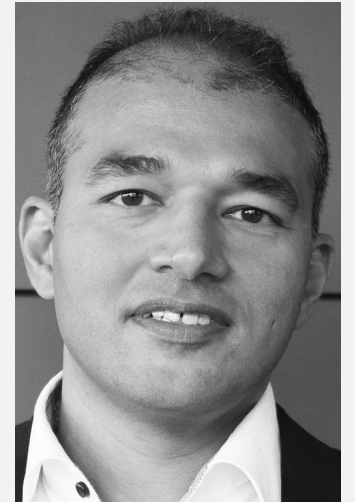
Human towers of Tarragona (The Guardian)



The course coordinator

Ali Muhammad – Assistant Professor at the SDU LSP Center

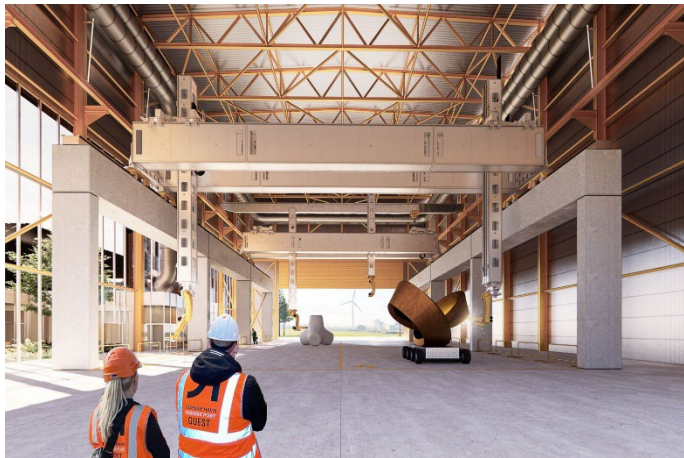
- Background
 - Located in Finland from 2001-2023
 - Moved to Denmark in 2024
 - Language proficiency: English (Fluent); Urdu (Native); Finnish (Basic); Arabic (Basic); Danish (Teach me)
- Education
 - MSc. in Automation and Control (2004) – Tampere University of Technology (Finland)
 - PhD. in Robotics (2011) – Tampere University of Technology (Finland)
- Professional
 - Principal Investigator in Robotics Systems at VTT National Research Centre of Finland (2011 – 2021)
 - Founder and consultant at Robotiq Pvt. Ltd. (2021 – 2023)
- Research Fields:
 - Embodied AI, Cognitive robotics, Multirobot systems
- Contact:
 - almu@mmmi.sdu.dk
 - Room: Ø27-602a-3
 - <https://portal.findresearcher.sdu.dk/en/persons/almu>





Centre of Large Structure Production

- Research sectors
 - Energy
 - Transport
 - Construction
- Research topics
 - Robotics
 - Control systems
 - Industry 4.0
- <https://largestructureproduction.sdu.dk/>



General rules

Any time you have a question, raise your hand and ask. It is ok to interrupt.

Do not talk/whisper during lecture. You may not realize, but you can be heard throughout the lecture room.

Do not use phones and other digital devices during the lecture. Overwhelming amount of research shows that they contribute negatively to the learning.



About the course

- System requirements
- Model and simulation-based design and development
- Prototyping and product development
- Lean product development methodology
- Business modelling
- Global sustainability goals
- Ethics, legals and social issues

About the course



- Application of engineering knowledge
- System development
- Rapid prototyping
- Market identification
- Business development
- Teamwork
- Pitching

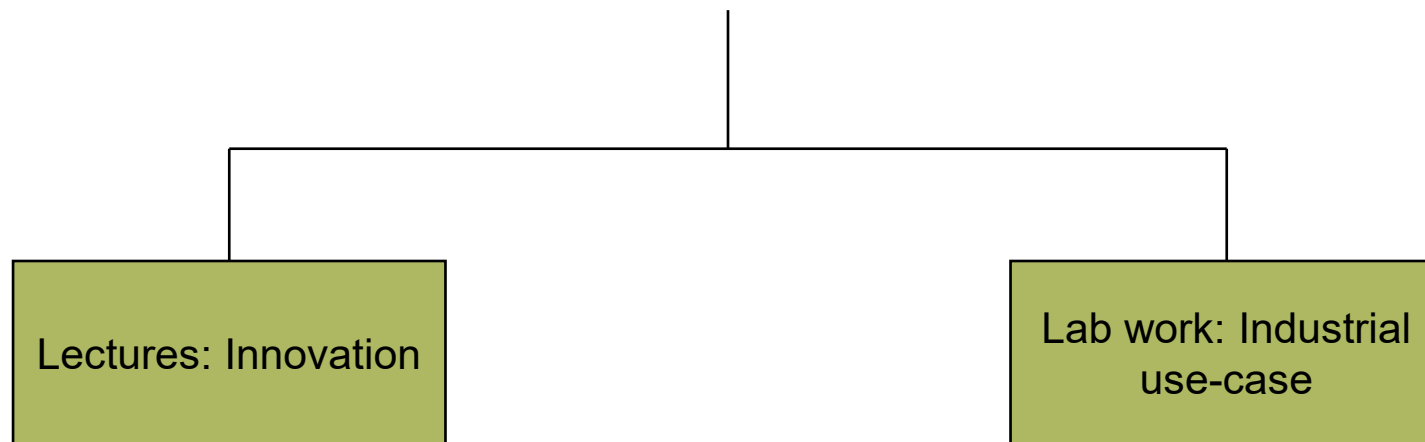
COMPETENCE



About the course

- Agile product development
- System understanding
- Lean startup methodology
- Self-reflection and constructive criticism

Course contents



Lecture topics



Time: Wednesdays 12:00 – 16:00
Room: U150

- User research
- Prototyping and user test
- Technology for sustainability (self study)
- Business modelling
- Ethics in technology (self study)
- Creating pitch
- Pitching

Some lecture sessions will be reserved for course information and providing feedback.

Lab work

- Robotics use cases (Dream lab)
- Drones use cases (Drone centre at H C Anderson airport)
- Software use cases (CoSE lab)



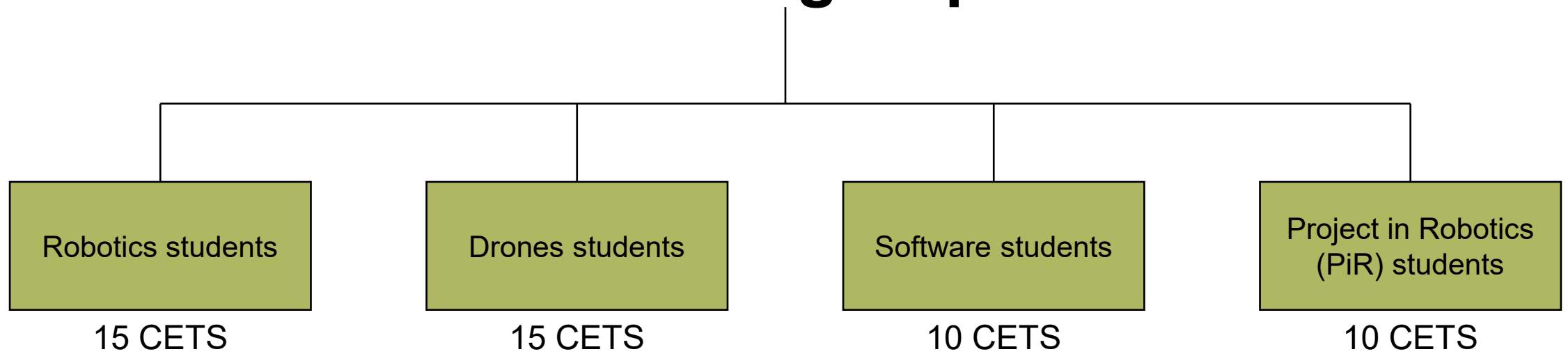
Time: Thursdays 8:00 – 16:00

Transport for Drone centre



Main door of TEK building at 8:00

Course groups



Course requirements

Software, Robotics and Drones students

- Attend lectures and apply new knowledge in the lab.
- Develop and demonstrate industrial challenge in the lab.
- Prepare a technical report on the demonstrator. Template will be provided.
- Prepare a pitch including video of the demonstrator. Template will be provided.
- Oral exam on **both reports**

PiR students

- Develop and demonstrate industrial challenge in the lab.
- Prepare a technical report on the demonstrator. Template will be provided.
- Presentation including video of demonstration
- Oral exam on **technical report**

Notes:

- Lab work, reports and presentation will be done in groups.
- Oral exam will be individual

Course requirements

How to prepare report on lectures

- Attend the lecture and do group work to see how the topic can be used in the lab for the development of industrial challenge.
- After the lecture: Describe the implication in your own industrial challenge (4 hours)

How to prepare technical report

- A template will be provided on Itslearning. 6-8 pages in the form of a scientific paper.

How to prepare pitch

- A template will be provided. It is a short presentation towards experts to convince them that your solution can create value.

How to prepare exam presentation

- Use pitch presentation and technical report as basis. Include more technical details and a video of the demonstration. Presentation time between 12-15 minutes. No template will be provided, so you are free to decide.

Automating the installation of cables



→ Challenge:

Many industrial and domestic products require the installation of wires, cables and harnesses. The robotic manipulation of flexible objects such as cables is challenging, as the object shape is not known exactly after grasping. Novel computer vision algorithms such as TrackDLO allow pose estimation of flexible objects (specifically Deformable Linear Objects (DLO)) and simulation software such as MuJoCo supports simulation of flexible objects. Consequently, learning-based policies for the manipulation of flexible objects can be developed based on state-of-the-art methods. Robotic manipulation of flexible objects may enable automation of tasks performed solely by humans today. This opens the possibility for new businesses within this field of application.

Automating the installation of cables



→ Challenge:

Autonomous mobile robots often use 2D or 3D lidar sensors for obstacle detection and navigation due to their precision and robustness in a variety of lighting conditions. However, lidar alone does not provide semantic understanding — it can detect that "something" is there, but not what that "something" is. RGB cameras, in contrast, are well-suited for object classification and semantic segmentation, but lack accurate depth information. Combining these two sensor types offers a promising way to enrich lidar data with meaningful object labels.

In real-world industrial settings, the robot frequently encounters semi-static objects such as chairs, pallets, bins, or carts — items that may move occasionally but often remain in place for hours or days. These objects should be handled differently than static structures like walls or pillars, which are used for localization. Without semantic understanding, robots may mistakenly treat all lidar returns as either static or dynamic, potentially corrupting maps or causing unnecessary avoidance behavior.

Automating the straightening of clothes



→ Challenge:

Straightening/folding clothes is an everyday domestic as well as industrial task. The robotic manipulation of flexible objects such as clothes is challenging, as the object shape is not known exactly after grasping. Artificial Intelligence models such as OpenVLA and startups like Physical Intelligence are demonstrating significant potential for robots to be able to perform such complicated tasks. Consequently, learning-based policies for the manipulation of flexible objects can be developed based on state-of-the-art methods. Robotic manipulation of flexible objects may enable automation of tasks performed solely by humans today. This opens the possibility for new businesses within this field of application.

Maximus D. Kaos

Associate Professor, Software Engineering

Co-Head, CHILL — Computer Human Interaction & Innovation Lab

Key focus areas:

- Technology for improving health-related quality of life
- Human-centered software engineering; playable, engaging systems
- Recognitions: Best Paper - CHI 2019; Academic Mindtrek 2021
- Keywords: Games, Play, Movement, HRQoL, XR, LLMs, AI

Background

PhD in Computer Science (2018). My work focuses on HCI to design, build, and evaluate interactive systems, especially XR experiences that support health, movement, and social connection.

Journey

SDU Odense (TEK) ← SDU Sønderborg (TEK) ← SDU Odense (Health)
← Aalto University (Finland) ← Queen's University (Canada)
← University of Arkansas (USA)



Contact

mkaos@mmmi.sdu.dk

Power Consumption of Robotic Solutions

Challenge: Develop a software tool to help developers and integrators improve robot efficiency by analyzing power consumption across hardware and infrastructure, enabling optimizations for both battery life and sustainability.

Scalable Computing for Robotic Solutions

Challenge: Develop a software method to help robotics companies reduce operational costs by optimizing the scalability and resource allocation for computationally intensive workloads like simulation and machine learning.





Developing a Multi-User VR System for Shared Physical Spaces

Challenge: Develop a comprehensive solution that demonstrates the integration of vacuum gripping technology with drones for payload pick and place operations.

Developing VR Training Applications with Force Feedback Gloves

Challenge: Develop an immersive VR training application to help technical or medical trainees master complex manual tasks by integrating force feedback gloves that provide realistic tactile sensations.



An optimization of the FC0 (FC1-FC5) calculation

Challenge: Develop a software tool to help DAO reduce operational overhead by radically optimizing their complex cost model calculation through algorithmic improvements and a more efficient coordinate system.

Optimization of address databases

Challenge: Develop a new, optimized address database to help DAO improve logistical efficiency by replacing a complex legacy system with a modern data structure that supports advanced route design and new business customers.



Codex-Agnosco: Local LLM-Assisted Legacy Code Analysis (VS Code Extension)

Challenge: Develop a privacy-first VS Code extension to help developers understand complex legacy code by using a local language model to generate explanations and documentation without compromising data security.

Database Migration Safety Analyzer

Challenge: Develop a static analysis tool to help DevOps teams and developers ensure database stability by proactively identifying high-risk changes in SQL migration scripts before they are deployed to production.

Digital Accessibility Compliance Checker

Challenge: Develop an automated auditing platform to help web and compliance teams improve digital inclusion by systematically identifying and reporting accessibility violations against legal standards like WCAG.

Robert Oliver Ladig

- Associate Professor at the SDU UAS Center (Dronecenter)
- Background
 - Located in Japan from 2011-2023
 - Moved to Denmark in 2023
 - Language proficiency:
 - German – Native
 - English – Fluent
 - Japanese – Proficient
 - Danish – _(ツ)_/
 - MS Degree – Robotics (2014) – Ritsumeikan University (Japan)
 - PhD Degree – Engineering (2018) – Ritsumeikan University (Japan)
 - Assistant Professor at the “Integrated Sensors and Intelligence Lab” at Ritsumeikan University (2018-2023)
 - Visiting Professor at “Center for Robot-Assisted Search and Rescue” at Texas A&M (2022)
- Research Fields and Fields of Interest:
 - Aerial Robotics, Aerial Manipulation, SAR Robotics, Computer Vision, XR, HMI
- Contact me
 - rola@mmmi.sdu.dk



Payload pick and place via drone



Challenge: Develop a comprehensive solution that demonstrates the integration of vacuum gripping technology with drones for payload pick and place operations.

Drone in a box



Challenge: Develop a comprehensive wildlife observation system utilizing the drone-in-a-box principle.

Objective 1 of today is to make sure that all students are part of the group.

Objective 2 of today is to make sure that all groups have the industrial challenge for the lab work tomorrow.

Let's build groups

We need to do little rearrangement. We need to divide the classroom in:

- Robotics students
 - Drone students
 - Software students
 - PiR students
- Please sit as groups together.
 - Students who do not have a group, join a group.

Software, Robotics and Drones students

- Robotics
 - 4 groups of 3 students
- Drones
 - 2 groups of 4 students
 - 3 groups of 5 students
- Software
 - 13 groups of 4 students
 - 1 group of 5 students

PiR students

- 2 groups of 3 students

Groups per industrial challenge

Category	Industrial challenge	Number of Teams
Robotics	Automating the installation of cables	2
	Semantic annotation of LIDAR data using RGB cameras	2
Drones	Payload pick and place via drone	2-3
	Done in a box	2-3
PiR	Automating the straightening of clothes	2

Groups per industrial challenge

Category	Industrial challenge	Number of Teams
Software	Power Consumption of Robotic Solutions	0-4
	Scalable Computing for Robotic Solutions	0-4
	Developing a Multi-User VR System for Shared Physical Spaces	0-4
	Developing VR Training Applications with Force Feedback Gloves	0-4
	An optimization of the FC0 (FC1-FC5) calculation	0-4
	Optimization of address databases	0-4
	Local LLM-Assisted Legacy Code Analysis (VS Code Extension)	0-4
	Database Migration Safety Analyzer	0-4
	Digital Accessibility Compliance Checker	0-4

Step 1: Choose a cool name for your group.

Step 2: Choose an industrial challenge for your group. Teachers will facilitate the process.

Step 3: Fill out the challenge selection table on my desk.

Step 4: Fill out and submit your group contract to me. Remember to keep a digital copy for yourself.

Group contract

- Name of the group: _____
- Title of the challenge: _____
- Name of the lab: _____
- Names of the group members:
 1. _____
 2. _____
 3. _____
 4. _____
 5. _____

**Tomorrow groups will be introduced to their
lab supervisors**

**Tomorrow groups will be introduced to their
allocated equipment**

Time: Thursdays 8:00 – 16:00